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Theoretical preparation

In the theoretical preparation of this exercise, we will primarily focus on the devices required to set up a network similar to the one in our assignment. There are four devices that we will encounter below:

Router

Switch

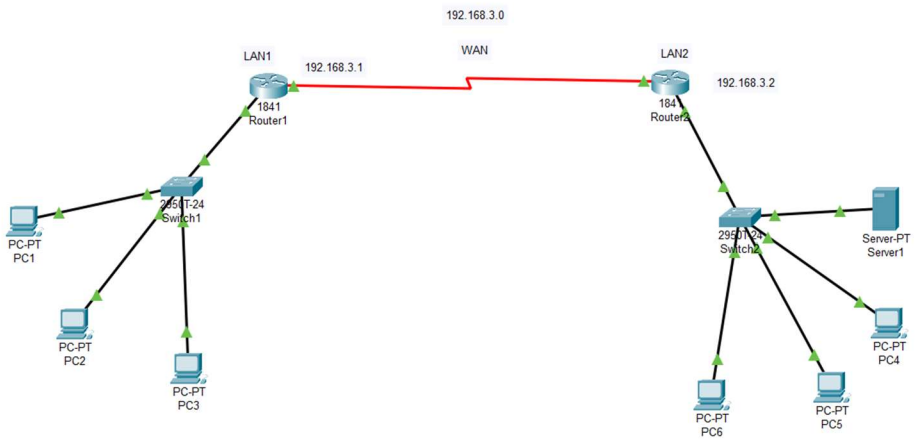
DHCP Server

Computer

Course of work

To begin with, we created the network by placing the devices in the emulator and connecting them using the cables suggested by the assignment. We then added modules to the router devices to be able to connect the two routers together. Immediately afterwards, we configured the WAN by assigning IP addresses and activating the clocking function on one router. The next step was to enable the connection between the routers and their respective switches. After successfully activating them, it was time to set IP addresses for our LANs. In LAN1, we had to assign static IPs manually as there is no DHCP server, unlike LAN2, where IP addresses are assigned dynamically. Finally, we enabled the WAN link, allowing the two routers to communicate with each other.

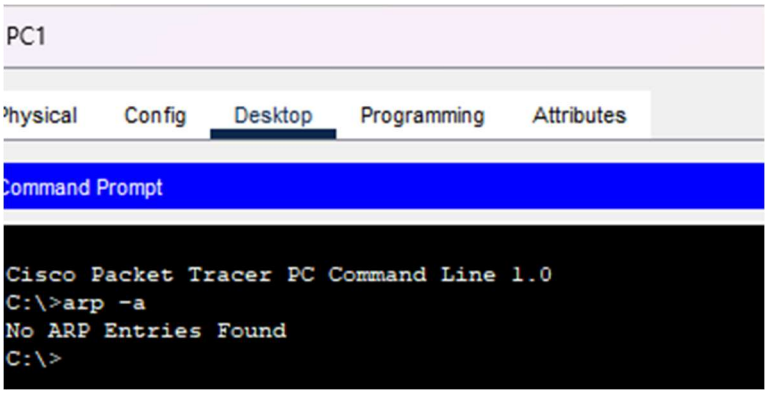
Screenshot


192.168.3.0
WAN
LAN1
192.168.3.1
1841
Router1
LAN2
192.168.3.2
Router2
2950T-24
Switch1
PC-PT
PC1
PC-PT
PC2
PC-PT
PC3
Server-PT
Server1
PC-PT
PC4
PC-PT
PC5
PC-PT
PC6

Answers to the questions

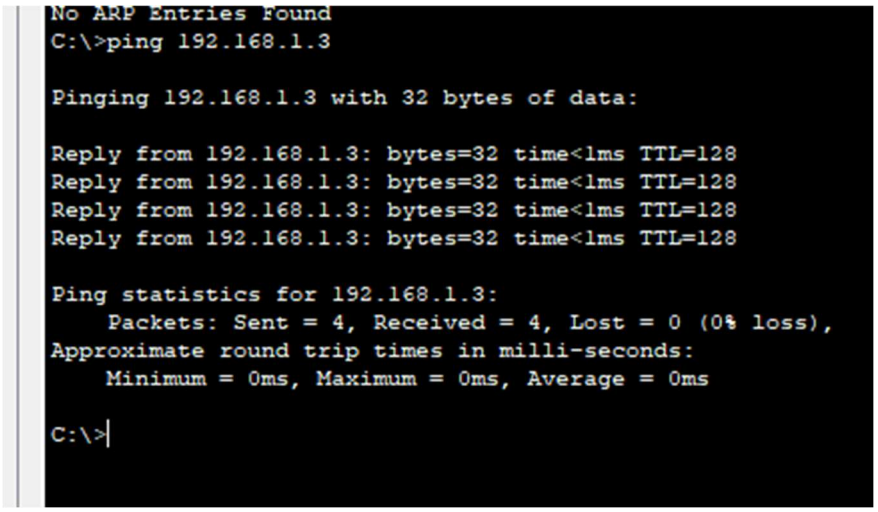
Step 6

In the Command Prompt on the PC1 Desktop with the command arp -a view the ARP table. What do you notice and why? (screenshot)



We notice that the ARP table is empty because PC1 has not communicated with any device so far.

Perform a ping from PC1 to PC2, PC3, PC4, PC5, and PC6. Is there a difference in the responses? (screenshot)



PC2 ping

```
C:\>ping 192.168.1.4

Pinging 192.168.1.4 with 32 bytes of data:

Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time=5ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 5ms, Average = 1ms

C:\>
```

PC3 ping

```
C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Reply from 192.168.2.2: bytes=32 time=11ms TTL=126
Reply from 192.168.2.2: bytes=32 time=7ms TTL=126
Reply from 192.168.2.2: bytes=32 time=7ms TTL=126
Reply from 192.168.2.2: bytes=32 time=8ms TTL=126

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 7ms, Maximum = 11ms, Average = 8ms

C:\>
```

PC4 ping

```
C:\>ping 192.168.2.3

Pinging 192.168.2.3 with 32 bytes of data:

Reply from 192.168.2.3: bytes=32 time=15ms TTL=126
Reply from 192.168.2.3: bytes=32 time=8ms TTL=126
Reply from 192.168.2.3: bytes=32 time=7ms TTL=126
Reply from 192.168.2.3: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.2.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 15ms, Average = 7ms

C:\>
```

PC5 ping

```
C:\>ping 192.168.2.4

Pinging 192.168.2.4 with 32 bytes of data:

Reply from 192.168.2.4: bytes=32 time=11ms TTL=126
Reply from 192.168.2.4: bytes=32 time=13ms TTL=126
Reply from 192.168.2.4: bytes=32 time=13ms TTL=126
Reply from 192.168.2.4: bytes=32 time=6ms TTL=126

Ping statistics for 192.168.2.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 6ms, Maximum = 13ms, Average = 10ms

C:\>
```

PC6 ping

As we can see, there is a difference in the time the packet travels when we ping a device from the other network (LAN2).

Look again at the ARP table on PC1. Is there a difference compared to Question 6? (screenshot)

```
C:\>arp -a
No ARP Entries Found
C:\>
```

Do any entries appear in the PC4 ARP table? Why?

PC4

Physical | Config | Desktop | Programming | Attribute

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>arp -a
No ARP Entries Found
C:\>
```

The reason nothing appears in either of the two tables is that Cisco Packet Tracer does not **retain the ARP cache** for devices that have communicated with each other in this simulation state.

Step 7

Using the command show ip route, confirm that the routers only "see" the LAN networks they are directly connected to(screenshot).

Router1

Physical Config CLI Attributes

IOS Command Line Interface

```
2 FastEthernet/IEEE 802.3 interface(s)
2 Low-speed serial(sync/async) network interface(s)
191K bytes of NVRAM.
63488K bytes of ATA CompactFlash (Read/Write)
Cisco IOS Software, 1841 Software (C1841-ADVIPSERVICESK9-M), Version 12.4(15)T1, RELEASE
SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2007 by Cisco Systems, Inc.
Compiled Wed 18-Jul-07 04:52 by pt_team

Press RETURN to get started!


%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up


Router>enable
Router#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

C    192.168.1.0/24 is directly connected, FastEthernet0/0
S    192.168.2.0/24 [1/0] via 192.168.3.2
C    192.168.3.0/24 is directly connected, Serial0/0/0

Router#
```

IOS Command Line Interface

```
2 FastEthernet/IEEE 802.3 interface(s)
2 Low-speed serial(sync/async) network interface(s)
191K bytes of NVRAM.
63488K bytes of ATA CompactFlash (Read/Write)
Cisco IOS Software, 1841 Software (C1841-ADVIPSERVICESK9-M), Version 12.4(15)T1, RELEASE
SOFTWARE (fc2)
Technical Support: http://www.cisco.com/techsupport
Copyright (c) 1986-2007 by Cisco Systems, Inc.
Compiled Wed 18-Jul-07 04:52 by pt_team

Press RETURN to get started!


%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

%LINK-5-CHANGED: Interface Serial0/0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0/0, changed state to up


Router>enable
Router#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

S    192.168.1.0/24 [1/0] via 192.168.3.1
C    192.168.2.0/24 is directly connected, FastEthernet0/0
C    192.168.3.0/24 is directly connected, Serial0/0/0

Router#
```

5. Ping from PC1 to PC2, PC3, PC4, PC5, and PC6. Do all PCs respond?

6. Ping from PC4 to PC1, PC2, PC3, PC5, and PC6. Do all PCs respond?

7. Export the ARP table from PC1 and PC4. Are the expected entries present?

As observed previously, all devices respond to the ping commands; however, the ARP tables remain empty for the reasons mentioned above.